Jonathan Parlett

Curriculum Vitae

jp86440@uga.edu
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Department of Mathematics
University of Georgia

Education

2025– **University of Georgia**
Ph.D. Mathematics, in progress

2020–2025 **Drexel University**
B.S. Mathematics; B.S. Computer Science
GPA 3.8

Research Interests

Arithmetic geometry, number theory, and combinatorics.

Publications

2025 Top to random and reverse: analysis of a new descent algebra shuffle, joint with Darij Grinberg. Preprint. [arXiv]

2025 Fixed Point Homing Shuffles. *Discrete Mathematics & Theoretical Computer Science*, vol. 27:1. [DOI]

2023 Subnet Communicability: Diffusive Communication Across the Brain Through a Backbone Subnetwork, joint with Abhishek Jeyapratap, Ali Shokoufandeh, Birkan Tunc, and Yusuf Osmanlioglu. *CDMRI 2023*. [Springer]

Talks and Posters

2025 Long Paths in Polynomial Divisor Graphs, joint with Nicole Froitzheim. Poster, Joint Mathematics Meetings. [poster]

2024 Long Paths in Polynomial Divisor Graphs, joint with Kayla Traxler. YMC 2024, Ohio State University. [slides]

2023 Subnet Communicability, joint with Abhishek Jeyapratap. CDMRI 2023. [slides]

Teaching

University of Georgia

Fall 2026 Instructor of record, Pre-calculus (Math 1113)

Fall 2026 Grader, Graduate Algebra I

Spring 2026 Grader, Introduction to Higher Mathematics

Fall 2025 Grader, Applied Linear Algebra

2020–2025 **Drexel University**
Tutor and grader in the computer science and mathematics departments, supporting data structures and algorithms, programming languages, algorithmic number theory and cryptography, and abstract and linear algebra. Responsibilities included holding office hours, grading, and occasionally creating supplemental course material.

Professional Experience

2022 **SRI International**, Princeton, NJ — Intern, data analysis and software engineering. Worked with Principal Research Engineer Norman Hurst to develop a technique to recover time from video streams, robust against frame rate conversions. Named co-inventor on the resulting patent.

2015–2020 **United States Marine Corps**, Camp Pendleton, CA — Data systems administrator. Configured and maintained information systems networks providing voice, email, file share, and internet services to facilitate communication between military units. Trained peers and subordinates.

Awards

Spring 2025 Robert J. Bickel Endowed Scholarship

Spring 2024 Outstanding Undergraduate Teaching Assistant Award

Spring 2024 Yilin Yang Outstanding Undergraduate Math Research Award

Winter 2023 Undergraduate Research Mini-Grant

2018, 2020 Navy and Marine Corps Achievement Medal, United States Marine Corps